

Andrea Valente

Bologna, Italy

[linkedin.com/in/andrea-valente0](https://www.linkedin.com/in/andrea-valente0) · github.com/Valendrew

AI computer vision engineer in applied R&D on open-ended client problems, from 3D geometry and custom deep learning models to the tools clients use to evaluate them. Traces issues to their root cause, whether in the data, the evaluation or the layers beneath the application.

Experience

Deep Vision Consulting

Modena, Italy

AI Computer Vision Engineer

Apr 2024 - Aug 2026

■ Automated package-seal inspection | global packaging manufacturer

- Worked with the client's sealing specialists, including on site, to turn how they judge torn package seals by hand into an interpretable inspection: physical force features, statistics of normal samples and computer vision (U-Net detection with RANSAC geometric fitting), combined so each covers cases the others miss while every result stays explainable. Adapted to three packaging materials, it matched or outperformed every specialist in the client's accuracy comparison.
- A critical seal defect was too rare to collect and could not be produced physically, so built a synthetic-data pipeline that turned a handful of annotated real images into thousands of realistic training images (thin-plate-spline warping) for a U-Net detector.
- Built the dashboard the client's sealing R&D team adopted as a working tool on new samples: training and evaluation jobs on a GPU queue, and each score shown beside its image overlays and force plots, so the specialists could see why a package failed and find pipeline problems themselves.

■ 3D face measurement from a single image | global luxury eyewear group

- The client wanted precise face measurements from a single phone or laptop image but had no way to measure 3D reconstruction accuracy. Built its first quantitative benchmark: fitted a 3D morphable head model to hand-annotated 3D face scans as ground truth, and scored reconstructions region by region with signed surface distances instead of one global error.
- Real 3D face data was scarce and rarely licensed for commercial use, so trained the reconstruction network on synthetic renders: a regressor of 3D face-shape parameters with a loss focused on key facial regions, across tracked experiments varying camera, pose, lighting and renderer, identifying the data and loss choices that worked best.
- Designed the protocol for a multi-device dataset with physical ground-truth measurements, validated and structured its several thousand captures, and built an interactive analysis tool the client used to explore accuracy and repeatability by device, pose, distance and lighting. The results showed single-image 3D was too ill-posed for the product, and the try-on kept its head-pose approach.

■ Head-pose model and in-browser inference for virtual try-on | global luxury eyewear group

- Designed and trained a compact network that predicts head position and rotation (6DoF) directly, giving the client a custom model it can extend and size per device instead of a landmark pipeline. The proposed backbone became the client's default at a quarter of the reference model's size, and was used in a pre-production version of the try-on.
- Designed the ground-truth strategy, combining pseudo-labels from the existing pipeline with synthetic faces whose landmarks are exact, and balanced the training mix across pose zones because the client's data was mostly frontal.
- Benchmarked the browser runtimes end to end across phones and browsers, then ported the C++/WebAssembly inference library to ONNX Runtime WebGPU for two models, with WGSL preprocessing shaders, browser tests and CI. It became the intended runtime of the pre-production try-on and was faster in the client's benchmark across low- to high-end phones.

Deep Vision Consulting | AI Computer Vision Engineer (continued)

- Investigated why a GPU preprocessing step took milliseconds when its GPU work took only microseconds: a Perfetto timeline across browser processes and the Chromium source showed that the browser checks for GPU completion on a timer, by design. Turned the finding into guidance for the pipeline: queue related GPU work, wait once, and overlap unavoidable waits with CPU work.

■ Shape fitting for scanned LED installations | smart decorative lighting company

- Fitted geometric shapes (boxes, cylinders, cones, pyramids) to scanned LED positions so lighting animations could follow an installation's form: a RANSAC framework with geometry-based candidates and a single bounded least-squares refinement, box fitting that accepts the few faces real installations show, and rejection of scans too incomplete to fit. Tested on synthetic and real point clouds.

AI Computer Vision Engineer (Intern)

Sep 2023 - Mar 2024

MSc thesis internship on 3D face reconstruction for virtual try-on.

- Compared multi-view reconstruction (neural radiance fields, multi-view stereo) with single-image 3D morphable-model regressors on 3D face scans, and chose the single-image route for a fast one-off face measurement, since multi-view methods needed dozens of images, precise camera poses and a still subject.
- Designed tens of thousands of synthetic training images from an analysis of how the head model's shape parameters change the face regions glasses rest on, and trained a shape regressor on them; its gap on real scans set the direction of the client project that followed: real ground truth by mesh fitting and a real benchmark.

Vaia Innovations

Remote, part-time

Frontend Web Developer

Jun 2022 - Sep 2022

- Built responsive Vue.js, TypeScript and Tailwind CSS views from Figma designs for article writing and exploration on Scripta, a Web3 publishing platform.

Skills

Computer vision and 3D: OpenCV, scikit-image, U-Net segmentation and keypoint detection, RANSAC and geometric fitting, 3D morphable head model fitting, head-pose estimation, point clouds (Open3D), PyTorch3D, Blender

Machine learning: PyTorch, timm, synthetic training data, model evaluation and benchmarking, MLflow

Inference and performance: ONNX Runtime, WebGPU (WGSL shaders), WebAssembly, TFLite, Perfetto profiling

Software engineering: Python, TypeScript, Docker, Celery, RabbitMQ, PostgreSQL, Azure, Playwright, CI, Git, Vue.js

Projects

Counterfactual explanations for a price classifier

GitHub · Demo · MSc, Mar - Jun 2023

- Two-person team: explained a PyTorch phone-price classifier by encoding it as mixed-integer constraints (OMLT, Pyomo, CPLEX), and compared the counterfactuals with DiCE's genetic search on validity, sparsity and realism; built the interactive FastAPI and Vue demo alone.

Rectangular circuit placement (VLSI)

GitHub · MSc, Jul - Sep 2022

- Two-person team: modelled the same strip-packing problem in constraint programming (MiniZinc), SMT (Z3, CVC4) and mixed-integer programming (PuLP, CPLEX) with a shared Python harness; the constraint-programming model solved all but one instance optimally, and a set-covering reformulation solved about a third more instances than the direct integer-programming model.

Comparative argument retrieval

GitHub · MSc, Nov 2022 - Jan 2023

- Three-person team: built sparse and dense indexes over a large passage corpus and compared keyword, dense, hybrid and MonoT5-reranked search for comparative questions; hybrid fusion gave the best recall and neural reranking the best top-five ranking.

Video context pipeline

GitHub · Personal, Sep 2026

- Designed and specified a typed asynchronous Python library that turns a video link into transcript, visual description, metadata and media through dependency-driven pipelines; tested it against the real transcription and video-understanding services, and released it as versioned GitHub releases.

Education

Master's Degree in Artificial Intelligence

Sep 2021 - Mar 2024

University of Bologna · 110/110 cum laude · Thesis: A study of 3D face reconstruction methods: from geometry to deformable mesh models

Bachelor's Degree in Computer Science

Sep 2018 - Dec 2021

University of Bologna · 110/110 cum laude · Thesis: evaluated a pretrained network for removing motion blur from X-ray tomographic images

Certifications and Training

International Computer Vision Summer School (ICVSS)

Jul 2025

Sicily, Italy · Selected by application while working in industry · Written exam passed, certificate

ELLIS Winter School on Foundation Models (FoMo)

Mar 2026

Amsterdam, Netherlands · Selected by application while working in industry · Attended

IELTS Academic, overall band 7.0

Jul 2021

Languages

Italian (native); English (C1, professional working proficiency)